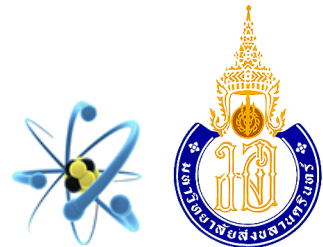


316-211

Ceramic Materials

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What is Ceramic Material ?



What is Ceramic Material ?



What is Ceramic Material ?



What is Ceramic Material ?



What is Ceramic Material ?



What is Ceramic Material ?



What is Ceramic Material ?



What is Ceramic Material ?

Which one is property of ceramics ?

Ceramic is a ductile material.

Ceramic is a brittle material.

What is Ceramic Material ?

Which one is property of ceramics ?

Ceramic consists metallic bonds.

Ceramic consist of ionic and/or covalent bonds.

What is Ceramic Material ?

Which one is property of ceramics ?

Ceramic can conduct electrical charges.

Ceramic can not conduct electrical charge.

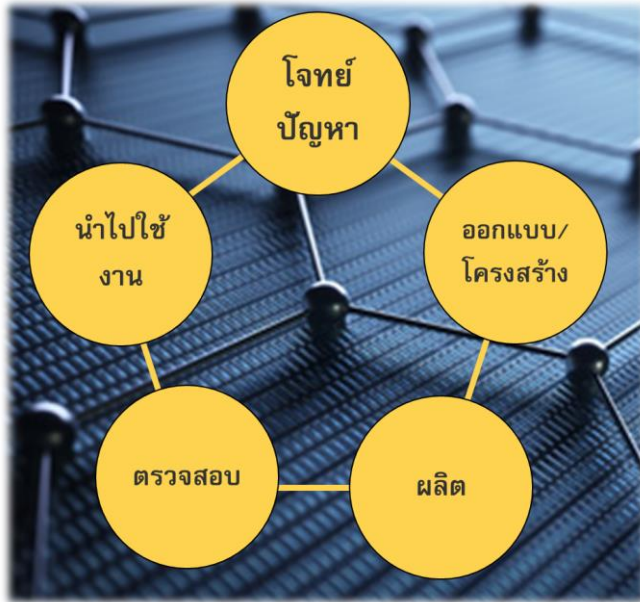
What is Ceramic Material ?

Which one is property of ceramics ?

Ceramic can be transparent / translucent.

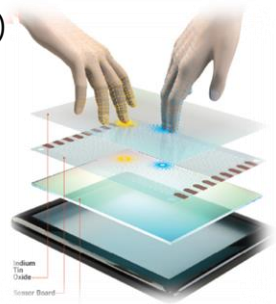
Ceramic can not be transparent / translucent.

What is Materials Science ?



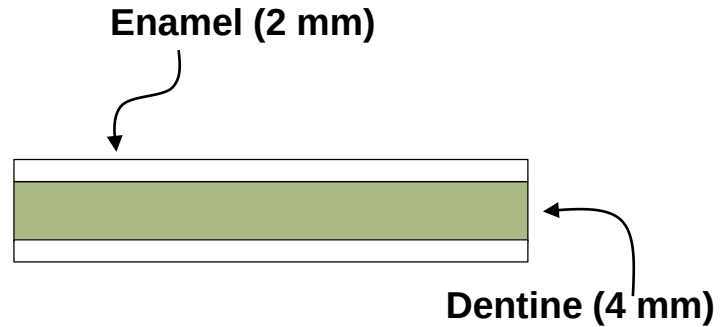
New Function Materials

- โลหะ (metals)
- เซรามิก (ceramics)
- พอลิเมอร์ (polymers)
- วัสดุผสม (composites)
- วัสดุนาโน (nanomats)



Examples of new function materials

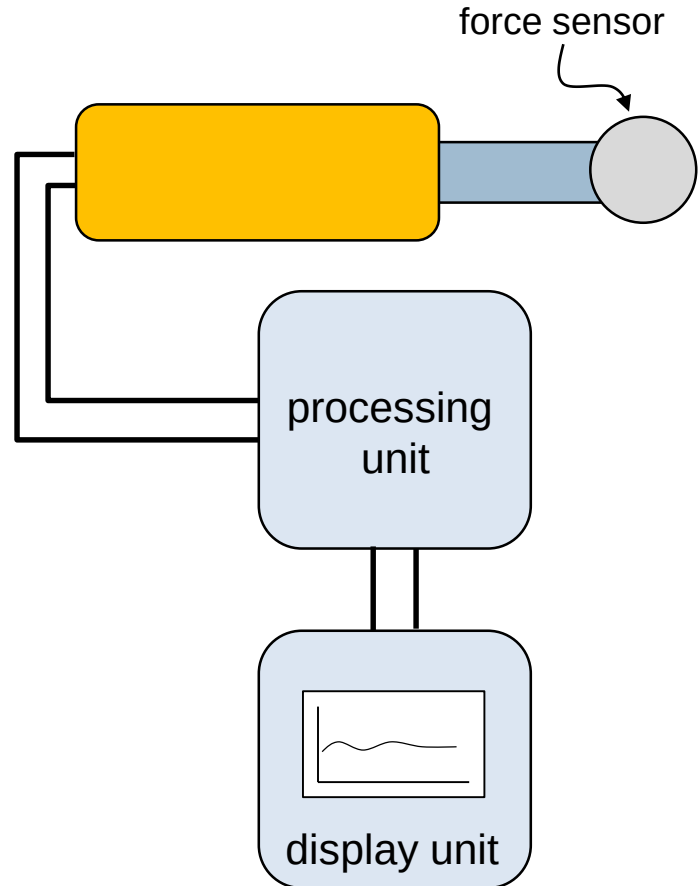
Composite grinding practice plate (senior project)



Cost: about 40 B/plate
Price: about 70 – 100 B/plate

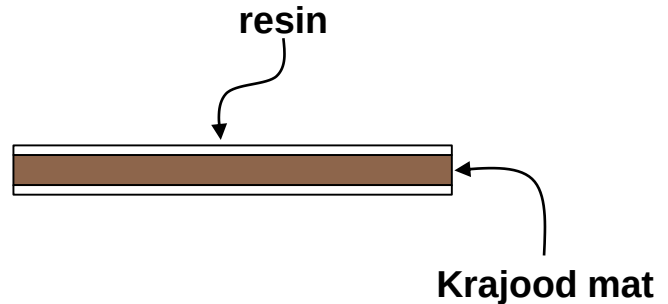
Examples of new function materials

Bite force sensor



Examples of new function materials

Composite wind turbine blade



Developing to be commercial products

1

Definition, History and Categories of ceramics



What does a ceramic mean ?

“A material that consists of **metallic and nonmetallic elements** bonding to each other with **ionic or semi-ionic/semi-covalent bonds** which behaves **high densities after firing**”







Development of ceramic materials

24,000 B.C.



Development of ceramic materials

14,000 B.C.



Development of ceramic materials

9000-10,000 B.C.



Development of ceramic materials

5000-8000 B.C.



Development of ceramic materials

1500 B.C.



Development of ceramic materials

1550 A.D.



Development of ceramic materials

Mid 1800's



Development of ceramic materials

1920's



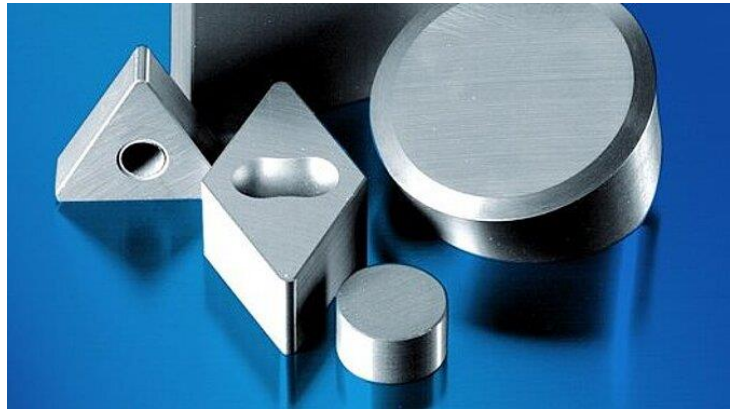
Development of ceramic materials

1940's



Development of ceramic materials

1960's



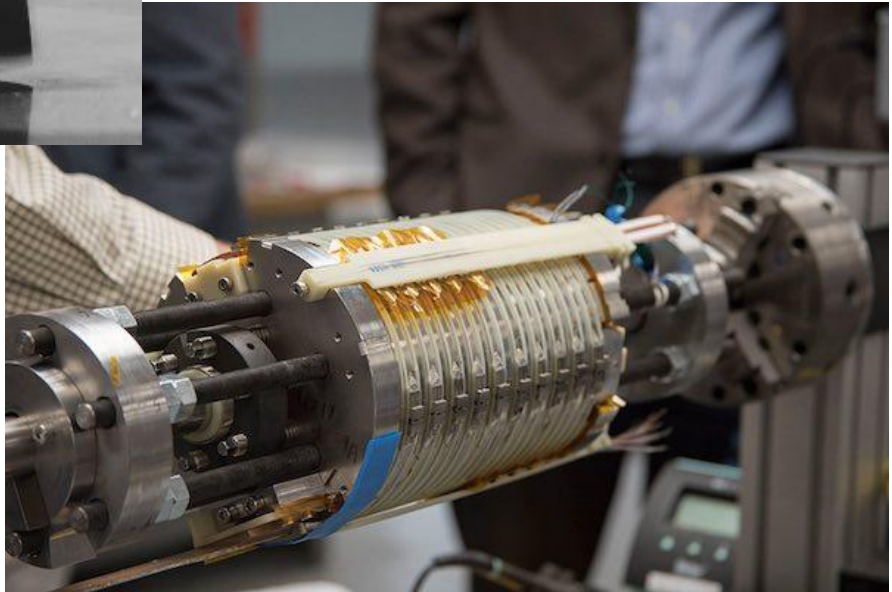
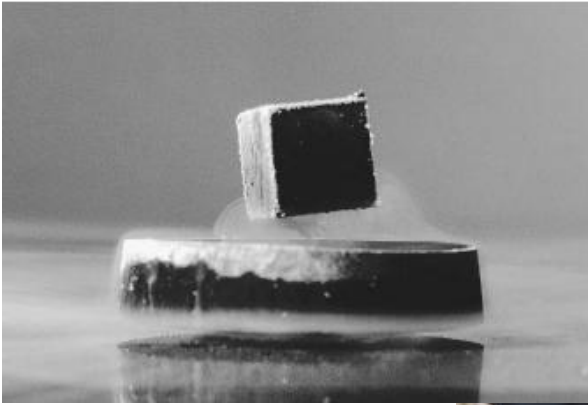
Development of ceramic materials

1970's



Development of ceramic materials

1980's

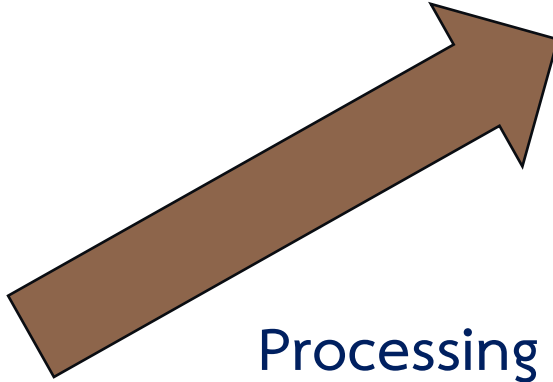


Development of ceramic materials

In the present, we want the ceramic that

- **makes human life faster, easier, cleaner and more convenient.**

Original of every changes



Applications



User

Processing



Materials Science



Materials Engineering

Objectives



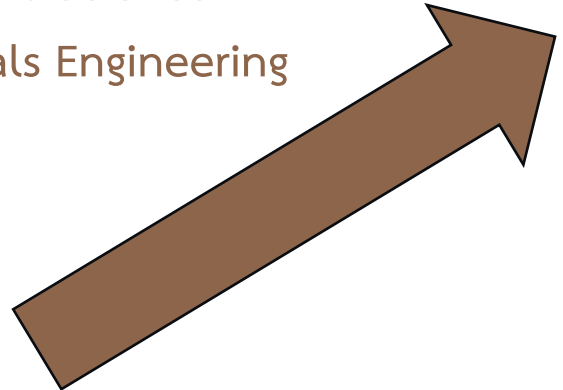
Problems (ปัญหา)



Necessary (ความจำเป็น)

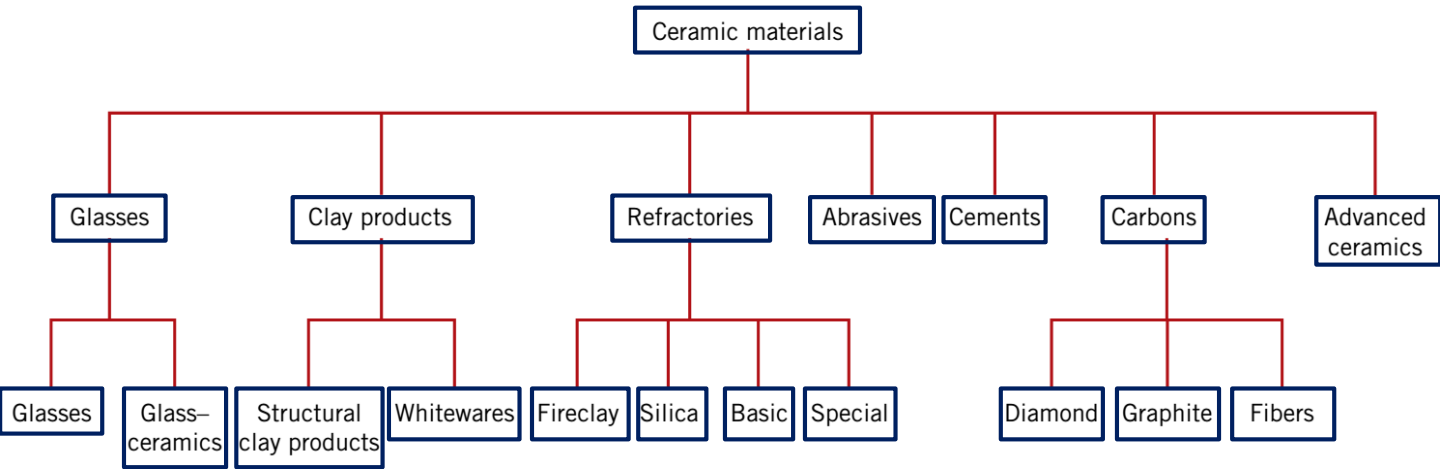


Want (ความต้องการ)



Categories of ceramic materials

Classification of ceramic materials on the **basis of application**



“Traditional and Advanced”

Glasses and glass ceramics



Structural clay products and whitewares



Refractories



Abrasive and Cements



Carbon, graphite and fibers



Advanced ceramics

